

The Universe Across Scales
IIIT-H, Semester: Winter 26, Module 2, Assignment 1

Submission deadline: April 10, 2026

1. Starting from the fluid equation (with $c = 1$)

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0,$$

show that for cold matter, for which $p = 0$, the energy density scales as $\rho \propto a^{-3}$.

2. Show that in a matter dominated universe (*i.e.*, $K = 0$, $\Lambda = 0$, and radiation energy density = 0) the scale factor varies with time as

$$a(t) = a_0 \frac{t^{2/3}}{t_0^{2/3}},$$

where a_0 is the value of the scale factor at present time $t = 0$.

3. Starting from spacetime metric show that the redshift and scale factors are related as

$$1 + z = \frac{a(t_0)}{a(t_e)}, \tag{1}$$

where $a(t_0)$ is the scale factor at the present time and t_e is the time the light ray was emitted.